**The devices used for holding the work in a drilling machine are**

1. Drill vise
2. T-bolt, nut and clamps step block
3. Plain vise
4. V – block
5. Angle plate
6. Drill jigs.

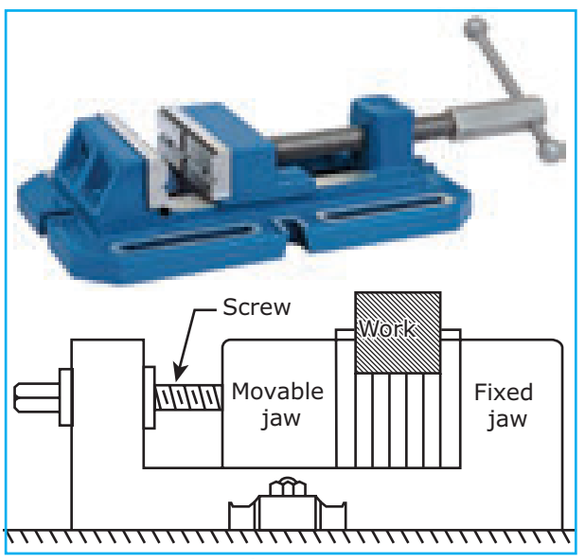
## Drill vise

Vise is one of the most important devices used for holding work piece on a drilling machine table. The work is clamped in a vise between a fixed jaw and a movable jaw.

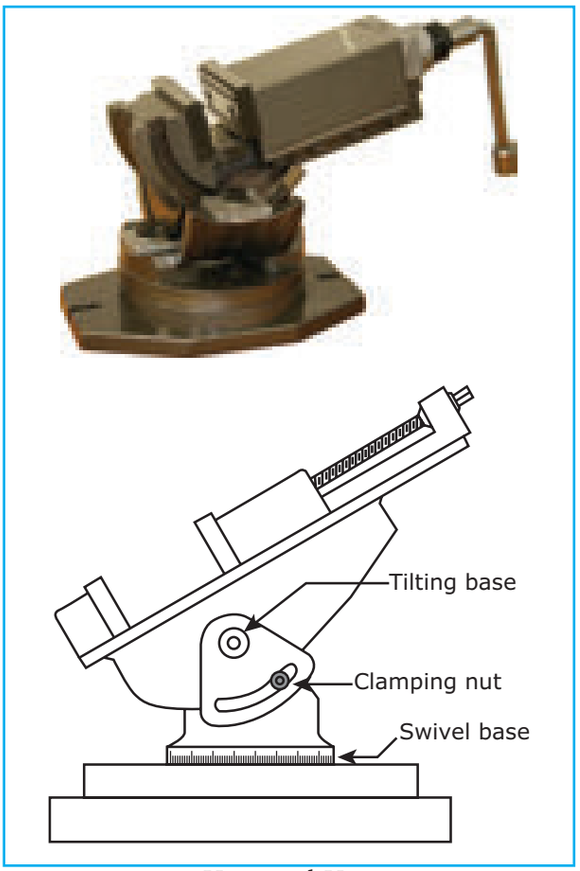
Parallel blocks are placed below the work. So that the drill may completely pass through the work without damaging the table. Different types of vises are used for holding different types of work and for performing different operations.

The different types of vises are

1. Plain vise
2. Swivel vise
3. Tilting vise
4. Universal vise



Plain Vise



Universal Vise

## T-bolts, clamps and step blocks

The work pieces can be held directly on the machine table by means of T-bolts clamps and step blocks. The top of the machine table has T-slots into which T-bolts may be fitted. The bolts of diameter 15 to 20 mm are used. The clamps and step – blocks are made of mild steel. T-bolts pass through a central hole on the clamp. The clamp is made to rest horizontally on the work surface by placing a suitable step block at the end of the work. Different height of the work pieces are held by levelling the clamp on different steps of the step block



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